

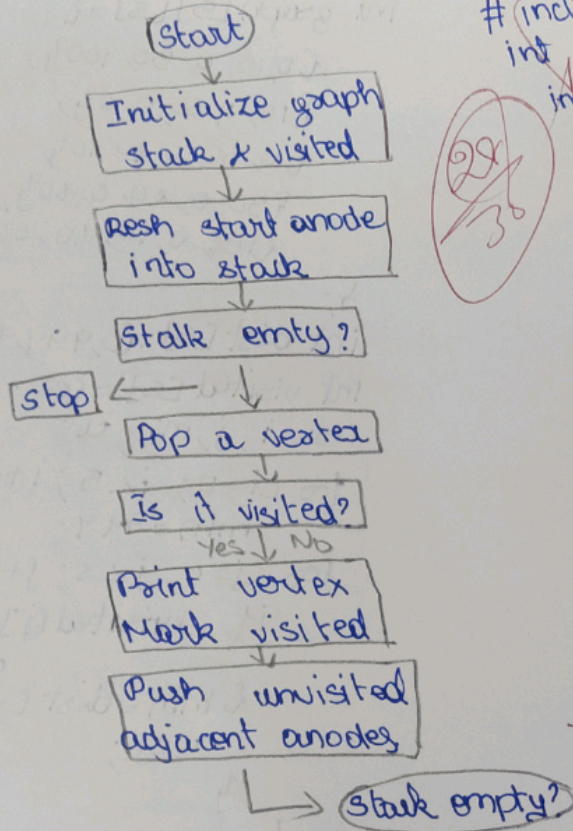
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CO5- AI3-Flowchart & Coding. DATA STRUCTURE - CSAD303.

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1. DFS using stack.

Flowchart



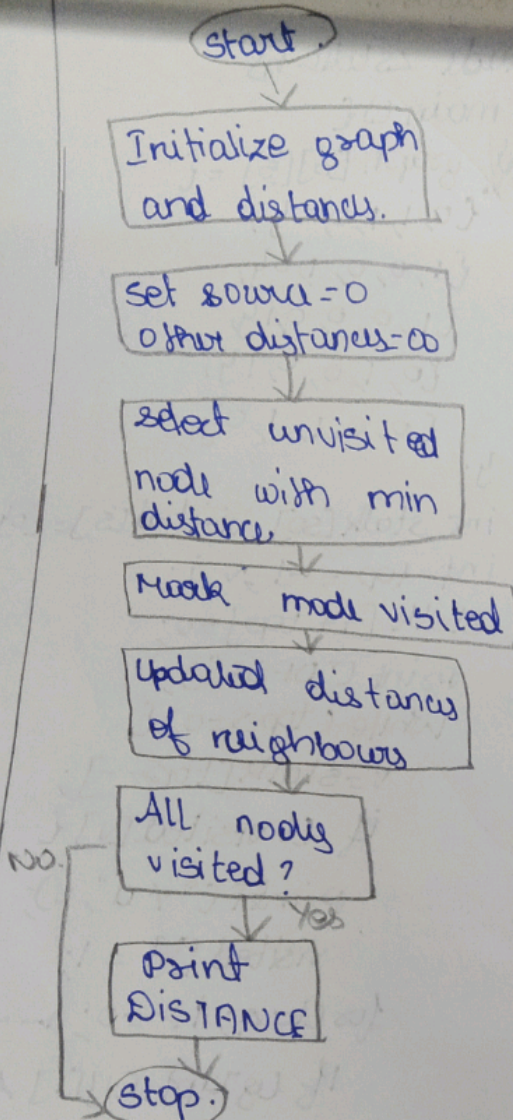
C-Program:

```

#include <stdio.h>
int main() {
    int graph[5][5] = {
        {0, 1, 1, 0, 0},
        {1, 0, 0, 1, 0},
        {1, 0, 0, 0, 1},
        {0, 1, 0, 0, 1},
        {0, 0, 1, 1, 0}
    };
    int stack[50], visited[5] = {0};
    int top = -1, n, i;
    stack[++top] = 0;
    printf("DFS:");
    while (top >= 0) {
        v = stack[top--];
        if (!visited[v]) {
            printf(" %d", v);
            visited[v] = 1;
            for (i = 4; i >= 0; i--)
                if (graph[v][i] && !visited[i])
                    stack[++top] = i;
        }
    }
    return 0;
}
  
```


2. Dijkstra's Algorithm.

Flowchart:



C-Program:

```
#include <stdio.h>
```

```
int main() {
```

```
    int graph[5][5] = {
```

```
        {0, 10, 0, 30, 100},
```

```
        {10, 0, 50, 0, 0},
```

```
        {0, 50, 0, 20, 10},
```

```
        {30, 0, 20, 0, 60},
```

```
        {100, 0, 10, 60, 0}}
```

```
};
```

```
int dist[5] = {0, 999, 999, 999, 999};
```

```
int visited[5] = {0};
```

```
int i, j, min, u;
```

```
for (i = 0; i < 5; i++) {
```

```
    min = 999;
```

```
    for (j = 0; j < 5; j++) {
```

```
        if (!visited[j] && dist[i]
```

```
            min).
```

```
        { min = dist[j];
```

```
        u = j;
```

```
    }
```

```
};
```

```
visited[u] = 1;
```

```
for (j = 0; j < 5; j++) {
```

```
    if (graph[u][j] != 0 &&
```

```
        dist[u] + graph[u][j] <
```

```
            dist[j])
```

```
        dist[j] = dist[u] + graph
```

```
        [u][j];
```

```
    }
```

```
for (i = 0; i < 5; i++)
```

```
    printf("0 -> %d = %d\n",
```

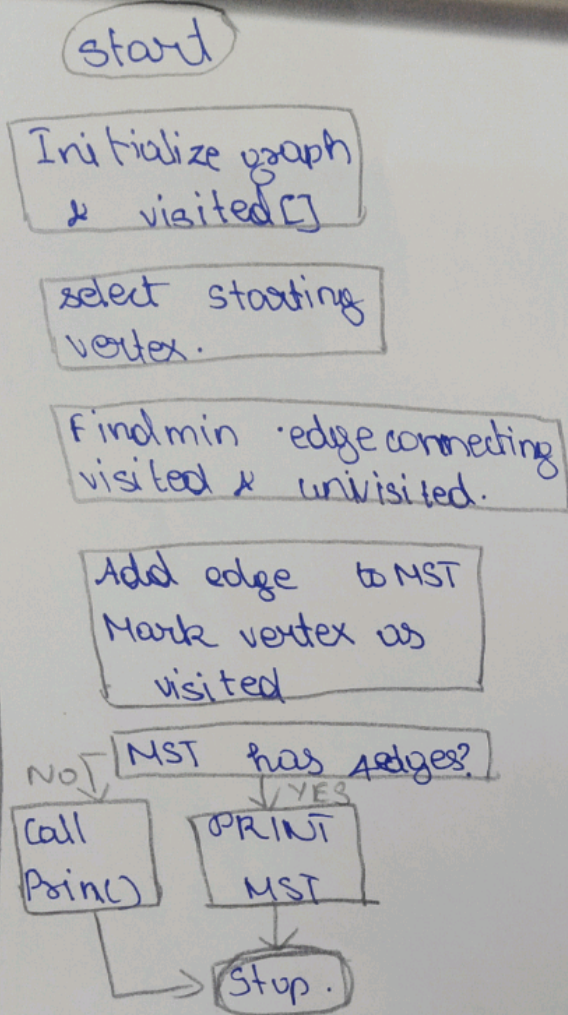
```
        i, dist[i]);
```

```
return 0;
```

```
}
```


Prim's Algorithm using Recursion;

Flowchart.



C-Program:

```

#include <stdio.h>
int graph[5][5] = {
    {0, 2, 0, 6, 0},
    {2, 0, 3, 8, 5},
    {0, 3, 0, 0, 7},
    {6, 8, 0, 0, 9},
    {0, 5, 7, 9, 0}
};

int visited[5] = {1, 0, 0, 0, 0};
void prim (int count) {
    int i, j, min = 999;
    int x = 0, y = 0;
    if (count == 4)
        return;
    for (i = 0; i < 5; i++) {
        if (visited[i]) {
            for (j = 0; j < 5; j++) {
                if (!visited[j] && graph[i][j] != 0 &&
                    graph[i][j] < min) {
                    min = graph[i][j];
                    x = i;
                    y = j;
                }
            }
        }
    }
    printf("\n x = %d y = %d min = %d",
           x, y, min);
    visited[y] = 1;
    prim(count + 1);
}

in main() { printf("Min Spanning Tree: \n"); prim(0); return 0; }
  
```